

Procedural Tree Property Reference

Shader Properties - Nature/Procedural Tree	
Property	What it does
_BarkMap	Sets the bark texture used on the trunk and branch geometry.
_BarkColor	Applies an overall color tint to the bark texture.
_BarkSmoothness	Controls how glossy or matte the bark surface appears.
_LeafMap	Sets the leaf texture used on the generated leaf cards.
_LeafColor	Applies an overall color tint to the leaf texture.
_LeafBottomTint	Colors the lower part of each leaf to add gradient variation.
_LeafTopTint	Colors the upper part of each leaf to create a lighter top gradient.
_Cutoff	Controls how much alpha is required before leaf pixels remain visible.
_LeafTranslucency	Adds backlighting so leaves glow more when lit from behind.
_LeafSmoothness	Controls how glossy or matte the leaf surface appears.
_WindStrength	Controls how far branches and leaves move in the wind.
_WindSpeed	Controls how fast the wind animation moves through the tree.
_WindScale	Controls the size and frequency of the wind wave pattern.

C# Fields - ProceduralTree.cs

Property	What it does
seed	Sets the random seed so the tree shape can be reproduced consistently.
treeHeight	Sets the overall height of the generated tree.
trunkRadius	Sets the base thickness of the trunk.
trunkLean	Controls how much the trunk can tilt away from vertical.
trunkCurvature	Controls how much the trunk bends along its path.
branchStartHeight	Sets how far up the trunk the main branches begin to appear.
radialSegments	Controls how many sides are used to build the trunk and branch tubes.
trunkSegments	Controls how many lengthwise segments are used for the trunk path.
branchSegments	Controls how many lengthwise segments are used for each branch path.
maxBranchDepth	Sets how many recursive branch levels can be generated.
mainBranchCountRange	Sets the minimum and maximum number of main branches spawned from the trunk.
childBranchCountRange	Sets the minimum and maximum number of child branches spawned from each branch.
mainBranchLengthRange	Sets the minimum and maximum relative length used for main branches.
childBranchLengthScale	Scales the length of child branches compared to their parent branch.
childBranchRadiusScale	Scales the thickness of child branches compared to their parent branch.
branchCurvature	Controls how much each branch bends as it grows.

C# Fields - ProceduralTree.cs

Property	What it does
branchUpwardBias	Controls how strongly branches bend upward while growing.
branchTwistJitter	Adds random angular variation to child branch directions.
minimumBranchLength	Sets the shortest branch length allowed before the branch stops growing.
minimumBranchRadius	Sets the thinnest branch radius allowed before the branch stops growing.
branchLeafCards	Sets the base number of leaf cards used on non-twig branch tips.
twigLeafCards	Sets the base number of leaf cards used on twig tips.
leafWidthRange	Sets the minimum and maximum width used for generated leaf cards.
leafHeightRange	Sets the minimum and maximum height used for generated leaf cards.
leafStartAlongBranch	Sets how far along a branch leaf anchors begin to appear.
leafClusterRadius	Controls the overall spread radius of each leaf cluster.
leafTiltJitter	Adds random tilt variation to generated leaf cards.
leafVerticalBias	Controls how much leaf clusters bias upward instead of spreading flat.
leafSeed	Sets the random seed used specifically for leaf placement and variation.
leafPositionVariance	Controls how much random positional variation is applied to leaf cards.
leafYawVariance	Controls the random side-to-side rotation variation of leaf cards.
leafRollVariance	Controls the random roll rotation variation of leaf cards.

C# Fields - ProceduralTree.cs

Property	What it does
leafRootPushIn	Pushes leaf roots inward so cards sit closer to the branch surface.
leafRootBackOffset	Offsets leaf roots backward along the branch direction for better attachment.
leafDirectionAlignment	Controls how strongly leaf card direction aligns with branch flow.
leafNormalAlignment	Controls how strongly leaf card facing aligns with the branch surface normal.
twigBranchCountRange	Sets the minimum and maximum number of twig branches that can spawn along a branch.
twigLengthRange	Sets the minimum and maximum relative length used for twig branches.
twigSegments	Controls how many lengthwise segments are used for each twig path.
twigRadiusScale	Scales twig thickness relative to the branch they grow from.
twigSpread	Controls the angular spread variation of generated twig branches.
leafAnchorDensity	Controls how many leaf anchor points are created along branches overall.
leafCardDensity	Controls the overall density multiplier used for leaf card generation.
terminalLeafDensity	Controls the leaf density multiplier applied to terminal tips.
twigLeafDensity	Controls the leaf density multiplier applied to twig tips.
barkUvScale	Controls the UV tiling scale of the bark texture along the tree.
useSingleSidedLeafGeometry	Chooses whether leaf cards use single-sided geometry instead of duplicated backfaces.
minimumRenderableBranchRadius	Sets the minimum branch radius required before full branch geometry is rendered.

C# Fields - ProceduralTree.cs

Property	What it does
minimumRenderableTwigRadius	Sets the minimum twig radius required before full twig geometry is rendered.
minimumLeafAnchorRadius	Sets the minimum radius required before a branch can generate leaf anchors.
minimumCapRadius	Sets the minimum radius required before branch ends receive a cap.
surfaceAnchorLeafCardScale	Scales the number of cards used by surface leaf anchors.
twigAnchorDensityScale	Scales how densely leaf anchors are generated on twig sources.
maximumRenderableBranchDepth	Sets the deepest branch level that can still render full branch geometry.
renderTwigGeometry	Enables or disables visible twig mesh generation.
surfaceAnchorDensityScale	Scales the density of surface-based leaf anchor generation.
terminalLeafCardDensityScale	Scales the card density applied to terminal leaf clusters.
terminalLeafSizeScale	Scales the size of leaf cards generated on terminal tips.
canopyClusterRadiusScale	Scales the spread radius of terminal canopy leaf clusters.
hiddenBranchLeafTipCount	Sets how many extra hidden-branch leaf tips are added along non-rendered branches.
hiddenBranchLeafDensity	Controls the density applied to leaves generated from hidden branches.
enableLods	Enables or disables automatic LOD mesh generation for the tree.
lod1Distance	Sets the distance where the tree switches from the full mesh to LOD1.
lod2Distance	Sets the distance where the tree switches from LOD1 to LOD2.

C# Fields - ProceduralTree.cs

Property	What it does
lodCullDistance	Sets the distance where the last LOD is no longer rendered.
lodReferenceFieldOfView	Sets the reference camera field of view used to calculate LOD transitions.
lod1Object	Stores the generated child object used for the first reduced-detail LOD mesh.
lod2Object	Stores the generated child object used for the second reduced-detail LOD mesh.